EXPERIMENT NO. 04

AIM: Build an app with buttons and gesture handling for event-driven interactions.

THEORY:

- **Event-driven programming**
  - Application responds to user actions like clicks and touches
  - Events trigger specific functions using listeners
- **Button Events**
  - `OnClickListener` handles button click events
  - `OnLongClickListener` handles long press actions
- **Gesture Handling**
  - Managed using `GestureDetector`
  - Detects gestures like:
    - Single tap
    - Double tap
    - Swipe (left/right/up/down)
- **User Interface Interaction**
  - UI components like Button and TextView respond dynamically
  - Provides interactive user experience

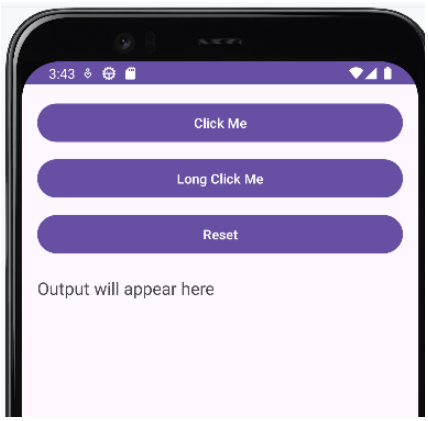
MainActivity.kt

```
package com.example.myapplication

import android.os.Bundle
import android.view.GestureDetector
import android.view.MotionEvent
import android.widget.Button
import android.widget.LinearLayout
import android.widget.TextView
import
androidx.appcompat.app.AppCompatActi
vity
import android.widget.Toast
class MainActivity :
AppCompatActivity() {

   private lateinit var
gestureDetector: GestureDetector
   private lateinit var tvOutput:
TextView

   override fun
onCreate(savedInstanceState:
Bundle?) {

super.onCreate(savedInstanceState)
setContentView(R.layout.activity_mai
n)

       val btnClick =
findViewById<Button>(R.id.btnClick)
       val btnLongClick =
findViewById<Button>(R.id.btnLongCli
ck)
       val layout =
findViewById<LinearLayout>(R.id.main
Layout)
       tvOutput =
findViewById(R.id.tvOutput)

       // Button click
       btnClick.setOnClickListener {
           tvOutput.text = "Button
Clicked"
           Toast.makeText(this,
"Button Clicked",
Toast.LENGTH_SHORT).show()
       }
```

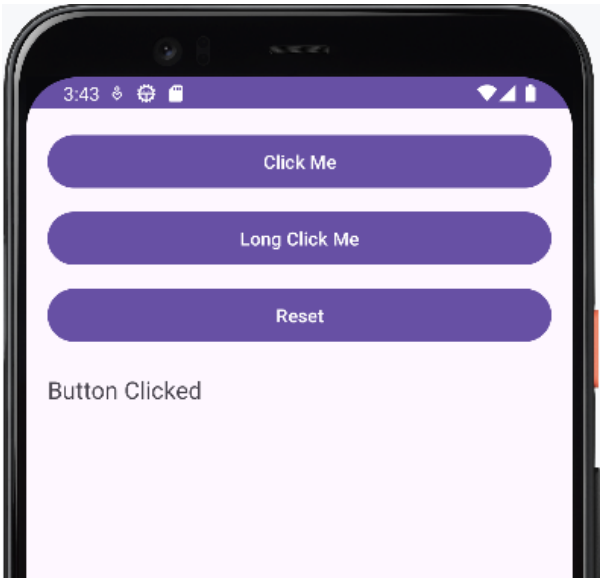
```
btnLongClick.setOnLongClickListener
{
            tvOutput.text = "Button
Long Clicked"
            Toast.makeText(this,
"Long Click",
Toast.LENGTH_SHORT).show()
            true
        }

        val btnReset =
findViewById<Button>(R.id.btnReset)

        btnReset.setOnClickListener {
            tvOutput.text = "Reset
Done"
        }

        // Gesture detector
        gestureDetector =
GestureDetector(this,
            object :
GestureDetector.SimpleOnGestureListe
ner() {

                override fun
onSingleTapConfirmed(e:
MotionEvent): Boolean {
                    tvOutput.text =
"Single Tap"

layout.setBackgroundColor(0xFFE1F5FE
.toInt()) // light blue
                    return true
                }

                override fun
onDoubleTap(e: MotionEvent): Boolean
{
                    tvOutput.text =
"Double Tap"

layout.setBackgroundColor(0xFFFFF9C4
.toInt()) // light yellow
                    return true
                }

                override fun onFling(
                    e1: MotionEvent?,
                    e2: MotionEvent,
                    velocityX: Float,
                    velocityY: Float
                ): Boolean {

                    val diffX = e2.x
- (e1?.x ?: 0f)
                    val diffY = e2.y
- (e1?.y ?: 0f)

                    if
(kotlin.math.abs(diffX) >
kotlin.math.abs(diffY)) {
                        if (diffX >
100) {

tvOutput.text = "Swipe Right"
                        } else if
(diffX < -100) {

tvOutput.text = "Swipe Left"
                        }
                    } else {
                        if (diffY >
100) {

tvOutput.text = "Swipe Down"
                        } else if
(diffY < -100) {

tvOutput.text = "Swipe Up"
                        }
                    }
                    return true
                }
            })
    }

    override fun onTouchEvent(event:
MotionEvent): Boolean {

gestureDetector.onTouchEvent(event)
        return
super.onTouchEvent(event)
    }
}
```

activity_main.xml

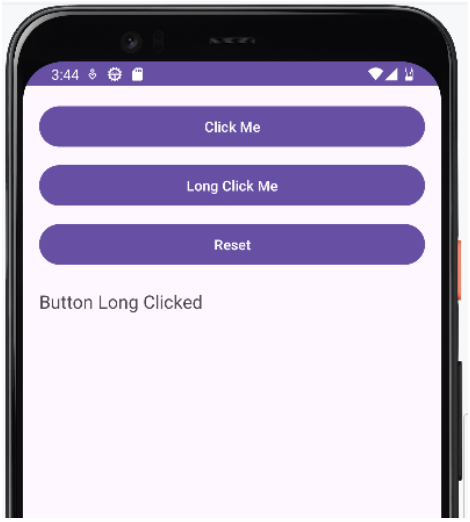
```
<?xml version="1.0"
encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.androi
d.com/apk/res/android"
   android:id="@+id/mainLayout"

android:layout_width="match_parent"

android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp">

   <Button
       android:id="@+id/btnClick"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Click Me"/>

   <Button

android:id="@+id/btnLongClick"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Long Click Me"

android:layout_marginTop="10dp"/>

   <Button
       android:id="@+id/btnReset"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Reset"

android:layout_marginTop="10dp"/>

   <TextView
       android:id="@+id/tvOutput"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Output will
appear here"
       android:textSize="18sp"

android:layout_marginTop="20dp"/>

</LinearLayout>
```



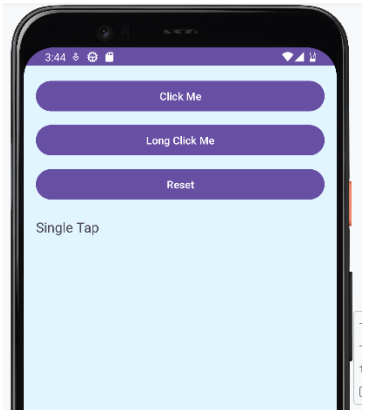
After clicking on click me



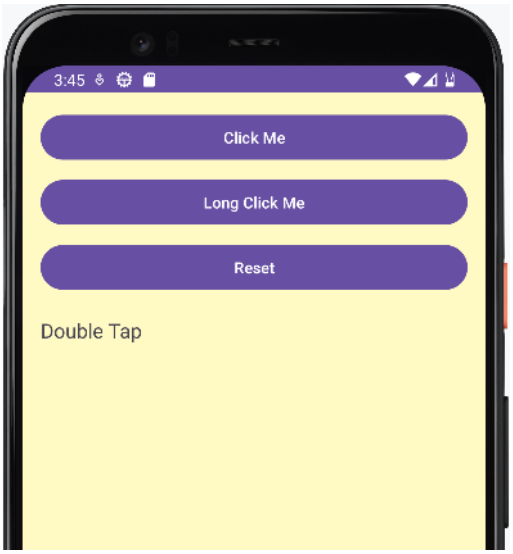
After long clicking 2nd button



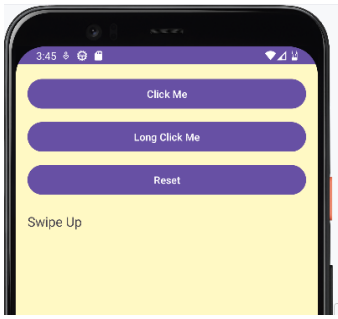
Single tap on screen changes background color



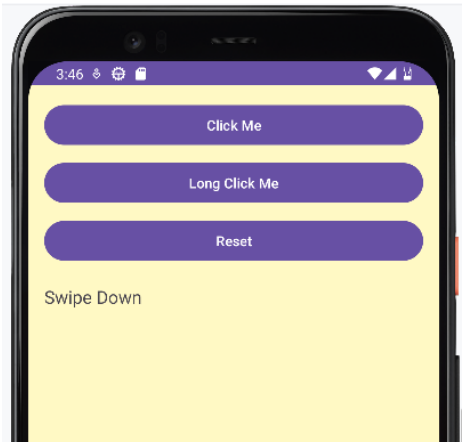
Double tap changes the background colour to yellow



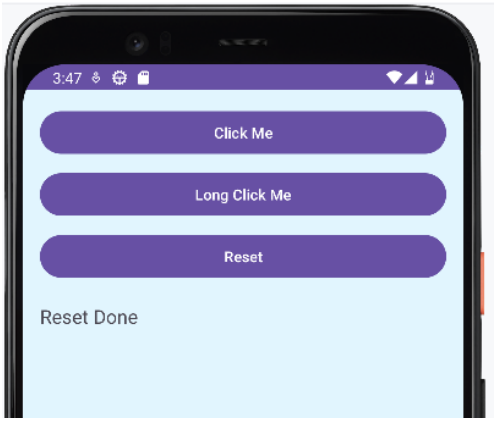
Message swipe up is displayed when user swipes upward



Message swipe down is displayed when user swipes downward



Clicking on rest button resets the settings



CONCLUSION:
This experiment demonstrates event-driven programming in Android by handling user interactions through button clicks and touch gestures, enhancing interactivity and responsiveness of the application.